Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–September 5th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure you provide a brief explanation. Remember, without a reason, the answer will be considered invalid.

A. What kind of synchronization does PRAM use? (1)

B. Which of the assumptions of the PRAM model simplifies the communication between the processor and the memory? (1)

C. Discuss the following sentence: "If a problem has no feasible/efficient solution on PRAM, it has no feasible/efficient solution for any parallel machine." True or False? (1)

D. Please compare the assumptions behind the Gustafson and the Amdahl laws. (1)

Exercise n. 2

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how the Divide and Conquer class of algorithms works by using one of the algorithms presented in class. (2)

B. Please explain what the competitive analysis is meant for by using one of the examples presented during the course lectures. (2)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain what the memoization technique is meant for and where it matters. (2)

B. Describe the Karger and Stein algorithm and its complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. What is meant by "accounting method"? To which class of algorithms could be applied? Examples? (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. Describe the skip list data structure and how the search is performed, and what the complexity is. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. Please explain what "Lower bound for comparison sorting" means and its implications. (1)